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THESIS PROPOSAL – MASTER OF INFORMATICS

Research Project Title	:	**Multi-Agent LLM Debate for Consensus-Based Difficulty Assessment in Automatic Playtesting**
Submitted by	:	**Muhammad Haydar Maulana**
Student ID	:	**203012510020**
Concentration	:	**Informatics Software Engineering**
E-mail address	:	**haydariday@gmail.com**
Supervisor (I)	:	**Dr. Arfive Gandhi S.T., M.T.I.**
Supervisor (II)	:	**Dr. Eng. Jati Hiliamsyah Husen S.T., M.Eng.**
Submitted Date	:	**20-06-2026**

Topic Summary

This research develops a multi-agent LLM debate framework for automated game difficulty assessment. Manual playtesting is resource-intensive and prohibitive for indie developers, while existing automated approaches—heuristics and deep learning—lack behavioral fidelity or require substantial computational resources. Recent single-agent LLM frameworks demonstrate cross-genre difficulty measurement capability but fail to capture the diversity of player play styles, achieving only limited accuracy when modeling heterogeneous personas.

The proposed framework operationalizes consensus-based difficulty assessment through structured multi-agent debate. A GM Agent proposes initial actions while Advisor Agents with distinct personas critique from opposing strategic perspectives, reaching consensus through iterative argumentation. This transforms difficulty measurement from single agent point estimates to aggregated multi-perspective assessments. The study employs Wordle and Slay the Spire as case studies.

Three research questions guide the investigation: (1) the framework's reliability across distinct game genres measured by outcome variance; (2) its comparative predictive validity against single LLM agents in terms of correlation with human performance baselines and computational cost; and (3) industry expert evaluations of practical utility and efficacy. Ground truth derives from empirical human performance data—guess counts for Wordle and remaining HP for Slay the Spire.

The research contributes methodologically by introducing automated consensus mechanisms for difficulty assessment via LLM debate; empirically by quantifying the performance-cost trade-off of multi-agent versus single-agent approaches; and practically by providing expert-validated guidelines for resource-constrained game development workflows.

Background

“Is the game hard enough for the intended player?” represents one of the key questions that haunting game designers. Balancing this problem risks losing player retention and possible negative reception

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[1], [2]. A playable game requires carefully calibrated difficulty—otherwise, it becomes stale, and players lose interest easily. That is why difficulty can be an indicator of player engagement and experience [3], [4]. To resolve this design tension, playtesting is conducted to help with measuring the difficulty. However, manual playtesting is a resource-intensive and rigorous process [1], [5], [6].

Automatic playtesting has witnessed gained popularity and is sought by industry and academic researchers alike. Various methods have been explored, ranging from traditional heuristics [7] to deep learning [5]. However, these approaches often require large training datasets and substantial computational resources, constraints that prove especially prohibitive for limited-budget companies such as indie game developers. Furthermore, while these approaches may outperform human players, they lack the consistency and natural behavioral pattern of actual human gameplay [8].

The emergence of Large Language Models (LLMs) provides alternative approaches for gameplay simulation and testing. Recent studies demonstrate that LLMs can simulate diverse games, ranging from board games such as chess [9] to complex games that require complex reasoning like Pokémon [10] and Minecraft [11]. Within difficulty measurement contexts, prior studies find that although LLMs lack full human-level capability, they can assess difficulty at levels comparable to human performance across distinct genres, making them viable as automated testers and quality assurance agents [12]. However, these existing multi-game frameworks do not account for the diversity of player play styles. Different play styles offer unique insights that can improve other styles [13]. Thus, capturing this diversity requires moving beyond single LLM agents. While multi-agent debate could reconcile these heterogeneous strategies into unified metrics, automated consensus frameworks remain absent from current difficulty assessment methods. However, whether such automated consensus mechanisms provide actionable insights for game designers and align with professional difficulty assessment practices remains an open question that requires expert validation.

This paper develops a multi-agent LLM debate consensus mechanism for automated difficulty assessment. We employ Wordle and Slay the Spire as case studies, which prior research [12] has established as well-designed games with generalizable structural features. While prior work theoretically establishes that multi-agent systems outperform single-agent configurations in modeling diverse behaviors [13], empirical validation within difficulty measurement contexts remains necessary. This comparison quantifies the incremental benefit of debate-based consensus in terms of correlation with human performance baselines and associated computational overhead. Expert opinion is further required to assess the practical utility and efficacy of the framework for professional game testing workflows.

Problem Identification

The research questions addressed in this paper are:

- **RQ1.** To what extent does the multi-agent debate framework produce reliable difficulty assessments across games (Wordle and Slay the Spire)?
- **RQ2.** How does the multi-agent framework compare to single LLM agents in terms of correlation with human performance baselines and computational cost?
- **RQ3.** How do industry experts evaluate the utility and efficacy of multi-agent LLM debate for automated difficulty testing?

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Objective and Hypotheses

The objectives of this research are threefold:

1. To implement a multi-agent LLM debate framework that operationalizes difficulty measurement through diverse personas, transforming point estimates into consensus-based assessments.
2. To evaluate the framework's generalizability across distinct game genres (Wordle and Slay the Spire) and validate its correlation with human performance baselines and computational cost against single LLM benchmarks.
3. To gather expert evaluations regarding the utility and efficacy of the multi-agent debate methodology for automated difficulty assessment in game development workflows

The hypothesis based on the RQs will be:

H1. The multi-agent debate framework produces reliable difficulty assessments across both Wordle and Slay the Spire, maintaining consensus convergence across game genres.
H2. The multi-agent framework achieves stronger correlation with human performance baselines than single LLM agents, albeit with higher computational.
H3. Industry experts will evaluate the multi-agent debate framework as viable for automated play tester.

Research Method

1. Reference Tracing

1.1. Difficulty And Its Impacts Towards Game

A "fun" game can be linked to the design of its difficulty. Players may feel a sense of accomplishment when overcoming challenges, particularly when those challenges are significant [2]. For example, Dark Souls was dubbed the hardest game of its time, yet the higher the risk the player must take, the greater the reward [14]. Defeating a worthy opponent gave the player a sense of greater enjoyment. The game garnered significant player engagement and attention—giving rise to a new game sub-genre known as “souls like” [15].

Given the importance of difficulty, measuring difficulty for the intended player is critical to game design. According to Constant et al. [16], there are two types of difficulty: First, subjective difficulty is how challenging players perceive the game to be. It cannot be measured explicitly because it is tied to personal player skill, stress, and emotion. Second, objective difficulty is a quantifiable, measurable aspect of the game that determines how challenging it is, independent of player perception. However, objective difficulty measurement requires rigorous handcrafted design by humans. While objective difficulty traditionally demands labor-intensive manual design, LLMs offer a promising alternative given their reasoning capabilities and human-like behavioral potential. Consequently, this study leverages LLMs to predict empirical difficulty metrics—specifically, human performance baseline data—automating the measurement of objective behavioral outcomes.

1.2. Automatic Playtesting

Alvarado Villa et al. [5]—utilizing the APLib framework—demonstrates a more "classical" yet highly powerful approach. Through goal structures and domain-specific languages (DSLs), they can create

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highly controlled test agents. However, this requires significant programming effort and a steep learning curve. Francillette et al. [7], with its mathematical model for platformer games, is also noteworthy. This model is highly deterministic and requires no sophisticated AI, but it is limited to the platformer genre and demands complex setup.

LLMs can indeed be employed to measure game difficulty, as demonstrated in Xiao et al. [12], despite their gameplay performance being inferior to human capability. The difficulty patterns experienced by LLMs resemble those perceived by humans. Nevertheless, for most games, single LLM performance proves inadequate, resulting in high loss rates and necessitating a handicap system to prevent this from occurring. This stems from the inherent limitations of single LLMs. Zhao et al. [6] adopts a different approach, focusing on match-3 games. They successfully address the LLM's inability to "see" by converting the game board into numerical representations. The results demonstrate strong code coverage and effective bug detection. Bateni et al. [17] exhibits superior long-term planning capabilities without specialized training, enabling adaptation to change in game rules.

2. Requirement Identification

2.1. Features

The system must do the following things. First, it must support multiple agents working together within a debate loop. Second, it must pull game state data from both Wordle and Slay the Spire and turn that structured data into text that the LLM can read. Third, the Agents must have different personas or strategic viewpoints so that the debate produces diverse opinions rather than everyone agreeing immediately. Fourth, the GM Agent must be able to stop the debate when enough perspectives have been heard and finalize the consensus action. Fifth, the system must translate the finalized consensus action into a format that the game engine can process and execute it directly within the game environment. Sixth, everything must be logged so that the difficulty correlation analysis can be done later.

2.2. LLM Integration

The framework needs a large language model that can handle complex reasoning and long conversations. At this stage, the research has not fixed the final model choice. The main contenders are DeepSeek and ChatGPT (OpenAI's GPT series), and the final pick will be whichever model represents the latest state-of-the-art when implementation begins. The chosen model must handle three basic things well. First, it must be good at multi-step reasoning, because the agents need to think through game states and argue back and forth. Second, it must accept long enough inputs to fit the full game description plus the debate history without cutting things off. Third, the model must be stable enough to run repeated experiments without constant failures. The cost of running the model will be tracked at the framework level so that RQ2 can compare whether the multi-agent approach is worth the extra expense.

2.3. Participants

For the expert evaluation in RQ3, the participants must be people who work in game development—specifically game designers, QA testers, or producers who have dealt with difficulty balancing or player experience. They need at least three years of industry experience, and it is preferred if they have worked on puzzle, strategy, or roguelike games like Wordle and Slay the Spire. The plan is to recruit five to seven experts through professional networks or game development communities. All participants will give informed consent, and their names and responses will be kept anonymous. This

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ensures that the evaluators know enough about real game development to judge whether the framework would be useful in practice.

3. Design Process

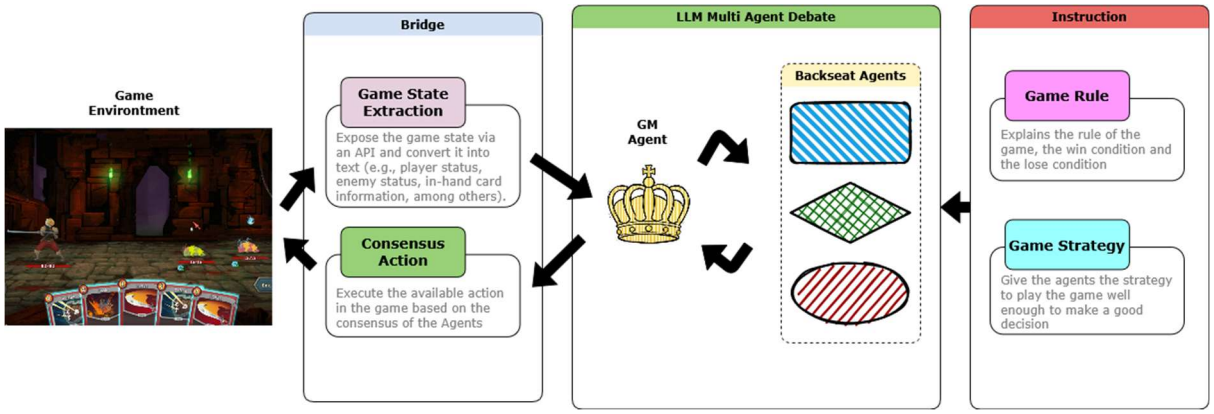


Figure 1 The multi-agent debate playtester framework

Figure 1 is the entire framework visualization. In each game loop, the game state is extracted and converted into text. Informed agents, equipped with game knowledge, make decisions beginning with the GM agent, who then relays its decision to backseat agents for debate. The GM agent makes the final decision via consensus, executing the action in the game.

3.1. Bridge

The Bridge serves as an **interface layer** connecting the game environment to the LLM multi-agent system. It consists of two sub-components:

3.1.1 Game State Extraction

- Retrieves raw game data from the game engine via API calls
- Converts structured game data (player status, enemy status, in-hand card information, among others) into textual representations that LLM agents can process

3.1.2 Consensus Action

- Executes the final action back into the game environment
- Receives the consensus decision from the GM Agent and translates it into game-engine commands (e.g., "play card X," "end turn," "choose path Y")

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3.2. Debate Protocol

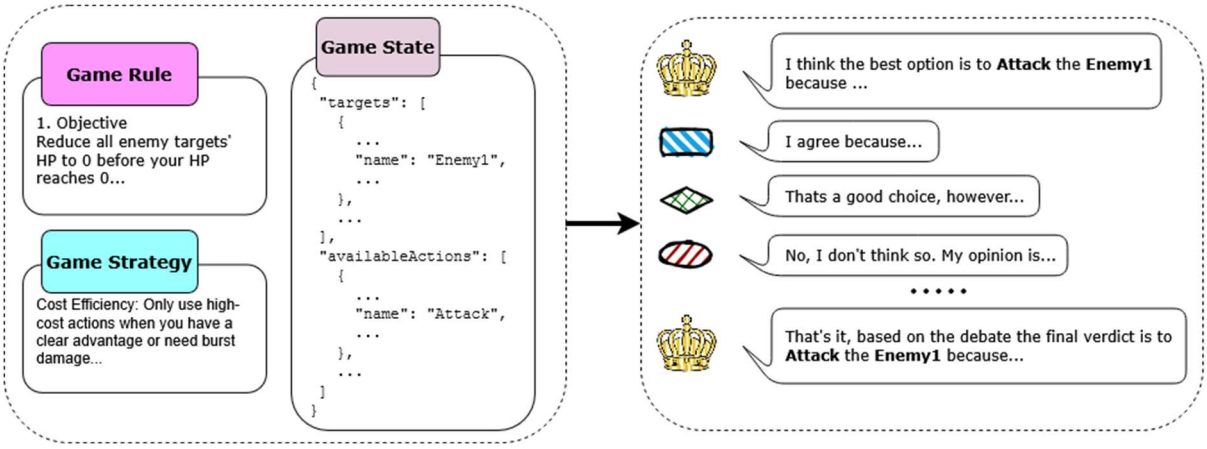


Figure 2 The In-depth Debate Protocol

The Debate Protocol governs how agents interact to reach action decisions based on the game state provided by the Bridge and the game knowledge from the Instruction module. Following Liang et al. [18], the protocol employs a structured multi-agent interaction where multiple agents express arguments in a state of "tit for tat"—a reciprocal exchange where each agent directly responds to the previous agent's position, challenging specific points and defending counterpoints in turn—while a managing agent oversees the process to obtain a final decision.

Figure 1 shows that the protocol has two distinct agent roles. First, the GM Agent receives the game state from the Bridge and proposes an initial action. This proposal is then relayed to the Advisor Agents, and after receiving their critiques, the GM Agent makes the final decision based on the accumulated debate. Second, the Advisor Agents also receive the game state, and they critique the GM Agent's proposal based on their distinct personas and strategic perspectives. They express agreement or disagreement with reasoned arguments and return their feedback to the GM Agent. These Advisor Agents have inherently different viewpoints which drive the debate and ensure that it captures diverse strategic considerations instead of just homogeneous agreement. The GM Agent's specific role is to initialize the argument and synthesize the final verdict, while all agents share equal access to the game state and knowledge modules.

The debate proceeds through iterative rounds; it can be looked in the Figure 2. In each round, the GM Agent presents its proposed action to the Advisor Agents. The Advisor Agents respond sequentially, challenging or supporting the proposal with reasoned arguments based on their interpretation of the game state and strategic considerations. The GM Agent revises its proposal in response to valid critiques, then determines whether further deliberation is productive or whether deliberation should conclude. This determination employs an adaptive break mechanism: rather than forcing a fixed number of debate rounds, the GM Agent evaluates whether additional iteration would yield meaningful change in the proposed action. If the GM Agent decides that enough perspectives have been considered and that further debate will not really change the decision, it applies to an adaptive break and finalizes the consensus action. If new, substantive critiques continue to emerge, the GM Agent initiates another round of debate until either convergence is achieved or a predefined maximum iteration limit is reached.

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3.3. Game Knowledge

The Instruction module provides static domain knowledge that is shared equally among all agents. This module consists of two main components. The first component is the Game Rules, which define the fundamental boundaries of gameplay. These rules include the winning conditions that agents must achieve, such as reducing enemy HP to zero, as well as the losing conditions that they must avoid, such as having their own HP reaching zero or running out of turns. The rules also specify the legal action spaces, which clarify what actions are permitted in any given game state. The second component is the Game Strategy, which supplies high-level heuristics that guide competent play. This strategic knowledge ensures that agents make informed decisions grounded in game understanding rather than selecting actions at random.

4. Implementation Process

4.1. Wordle

Wordle is a word-guessing game where players attempt to identify a hidden five-letter target word within six attempts. After each guess, the game provides feedback indicating which letters are correct and in the right position (green), correct but in the wrong position (yellow), or not in the word at all (gray). The Wordle implementation is built as a lightweight Python prototype with a custom engine that simulates these core game mechanics, processes guess inputs and returns standard feedback patterns. A custom Bridge module connects this prototype to the LLM agents by polling the game state after each guess and converting it into text that includes guess history, feedback patterns, remaining attempts, and confirmed or eliminated letters. The Bridge also validates incoming consensus guesses before submitting them to the game. A custom Instruction module supplies all agents with static domain knowledge, including game rules such as five-letter constraints and green-yellow-gray feedback conventions, plus high-level strategy heuristics like prioritizing high-frequency letters and narrowing the solution space efficiently.

The debate structure for Wordle consists of one GM Agent and two Advisor Agents with opposing roles. After the GM Agent proposes an initial guess word based on the current game state, the **Negative Agent** opposes this proposal by identifying weaknesses and presenting an alternative guess with supporting reasoning. The **Positive Agent** then opposes the Negative Agent's critique by defending the GM's original proposal, highlighting its strengths and explaining why the alternative may be inferior. This creates a structured back-and-forth where the GM's initial argument is tested against direct opposition and then defended by a counter-opposing voice.

Once both Advisor Agents have spoken, the GM Agent reviews the full debate history and makes one of two decisions. It may conclude that sufficient consensus has been reached and finalize the guess word, either sticking with its original proposal or revising it based on the arguments presented. Alternatively, it may determine that significant disagreement remains and allow the Negative Agent to present an additional argument based on the updated debate history. The debate is currently capped at three rounds maximum for this prototype, though this limit is configurable and may be adjusted during the actual implementation phase; once the limit is reached or the GM decides to end early, the final guess is passed through the Bridge to the Wordle prototype for execution, and the resulting feedback becomes the new game state for the next turn.

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4.2. Slay The Spire

Slay the Spire is a deck-building roguelike where players navigate combat encounters using a hand of cards that grant attacks, defenses, and special abilities, with the goal of reducing enemy HP to zero before their own HP is depleted. For this implementation, the game is modified with two essential mods: BasicMod, which serves as the fundamental infrastructure required for other mods to operate, and CommunicationMod, which enables the extraction of real-time game information and the submission of input actions through standard streams (stdin/stdout). CommunicationMod also provides a Python interface, allowing the Bridge module to interact with the game through APIs rather than building a custom engine from scratch. This means the Bridge can directly poll the game state and inject the consensus action back into the game without screen parsing or memory reading.

The debate structure for Slay the Spire follows the same three-agent framework as Wordle but expands to three Advisor Agents with distinct combat personalities. The Aggressive Agent evaluates proposals solely based on how quickly and effectively the enemy can be killed, prioritizing damage output above all else. The Defensive Agent opposes this by emphasizing survivability, arguing for actions that maximize block, healing, or risk mitigation to preserve player HP. The Velocity Agent takes a longer-term perspective, focusing on card draw efficiency, energy management, and whether an action sets up powerful combo sequences for future turns rather than just immediate impact. These three personas ensure that every proposed action is stress-tested against competing strategic priorities before reaching consensus.

The debate flow mirrors the Wordle implementation: the GM Agent proposes an initial action based on the current combat state, then the three Advisor Agents respond in fixed sequence—first Aggressive, then Defensive, then Velocity—each critiquing or supporting the proposal from their specialized viewpoint. After all three have spoken, the GM Agent reviews the debate history and either finalizes the consensus action or allows one additional round of critique if significant disagreement remains. The debate is capped at three rounds maximum, though this limit is configurable and may be adjusted during the actual implementation phase. Once concluded, the final action is passed through the Bridge to the CommunicationMod API for execution, and the resulting game state is fed back into the loop. The fixed sequence of Aggressive, Defensive, and Velocity represents the current project scope; more dynamic ordering or adaptive persona selection is reserved for future work.

5. Experiment Design and Plan

The experiment employs a mixed-methods approach. Quantitative evaluation addresses RQ1 and RQ2 through controlled comparison across two game environments, while qualitative evaluation addresses RQ3 through structured expert assessment. The multi-agent framework is tested against a single-agent baseline under identical conditions to isolate the effect of the debate mechanism.

In Wordle, the difficulty is operationalized through the total guess count proxy—how many attempts a player requires to identify the target word. Lower guess counts indicate easier words, on the other hand higher counts indicate harder words. A sample of target words is selected spanning known difficulty tiers. While Slay the Spire**,** the difficulty is operationalized through the remaining health points (HP) proxy at the conclusion of a combat encounter. Higher remaining HP indicates an easier encounter, on the other hand lower or zero remaining HP indicates a harder encounter. Encounter configurations are selected across the boss of Acts 1 & 2.

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For both games, the ground truth is derived from empirical human performance data rather than subjective ratings. For Wordle, the dataset comes from engaging-data.com[1], which provides gameplay samples from real human players along with the average number of steps taken for each puzzle. For example, a straightforward mapping where a word with an average of 3.2 guesses is objectively easier than one with 5.8 guesses. However, the dataset does not provide additional statistical details such as demographic information about the players. For Slay the Spire, the developers have released a large-scale dataset containing over 75 million recorded human gameplay sessions. From this, the average remaining HP per encounter configuration can be extracted. For example, establishing that an encounter leaving players with 80 HP on average is objectively easier than one leaving 10 HP. These empirical baselines represent the human performance proxy that the LLM frameworks attempt to predict.

For RQ1, reliability is assessed through outcome variance: the framework is run repeatedly on identical scenarios and the standard deviation of playthrough results is calculated, with low variance indicating that the debate reliably converges on similar outcomes. This metric is compared across both games to test whether the mechanism remains stable. For RQ2, predictive validity is measured by the Pearson correlation between the framework's playthrough outcomes and the human performance baseline, calculated separately for multi-agent and single-agent modes to determine which better predicts actual human difficulty, while computational cost is quantified as a token ratio dividing multi-agent consumption by single-agent consumption. For RQ3, expert evaluation combines a Likert scale survey where professionals rate utility statements on a 1-to-5 scale and the overall average score is reported, alongside a thematic analysis of semi-structured interviews that identifies recurring themes and reports the frequency of each theme as a percentage of the expert pool.

[1] https://engaging-data.com/wordle-guess-distribution/

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Timetable/Schedule

Activity / Sub-Activity		1	2	3	4	5	6	7	8	9	10	11	12
Reference tracing	Automatic playtesting	■	■										
	Game difficulty	■	■										
	Gap identification		■	■									
Requirement Identificaion	Eval: LLM and expert profile	■	■										
	RQs, hypotheses, and scope		■	■									
Design process	Bridge	■	■										
	Debate protocol		■	■									
	Data logging			■	■								
Implementation process	Wordle prototype		■	■	■	■							
	Slay the Spire development			■	■	■	■						
Experiment design and plan	Ground truth data acquisition	■	■										
	RQ1 and RQ2			■	■	■	■						
	RQ3							■	■				
Analysis/Evaluation	Quantitative				■	■	■	■	■				
	Expert evaluation							■	■	■			
	Thematic analysis								■	■			
Paper Submission	Drafting				■	■	■						
	Internal revision							■	■	■			
	Conference/journal submission										■	■	■

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Supervisor (I)'s Comments:

Comments about the title

OK

Comments about the research method

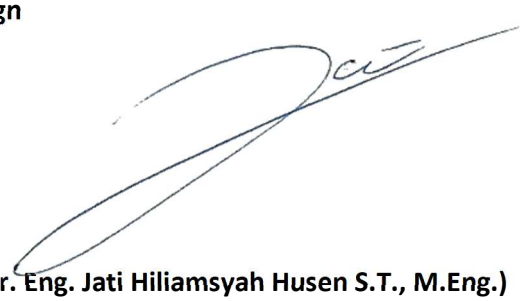
Riset perlu memperhatikan permasalahan awalnya dan apakah permasalahan akan terjawab dengan riset ini. Scopenya juga perlu diperhatikan karena riset hanya 1 tahun kurang. Perlu juga lebih diperjelas bagaimana seluruh agent mencapai consensus.

Sign **Date: 19-06-2026**

(Dr. Arfive Gandhi S.T., M.T.I.)

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Supervisor (II)'s Comments:

Comments about the title **OK**
Comments about the research method **Pada implementasi harus memiliki bukti dan alasan mengapa memilih debat competition, berikan alasannya juga. Saran untuk memilih dataset yang banyak bisa menggunakan random sample karena untuk mengecek data yang bersifat populasi, bukan individu.**
Sign **Date: 19-06-2026**      **(Dr. Eng. Jati Hiliamsyah Husen S.T., M.Eng.)**

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Reference

[1] M. Weber dan P. Notargiacomo, "Dynamic Difficulty Adjustment in Digital Games Using Genetic Algorithms," dalam *2020 19th Brazilian Symposium on Computer Games and Digital Entertainment (SBGames)*, Recife, Brazil: IEEE, Nov 2020, hlm. 62–70. doi: 10.1109/SBGames51465.2020.00019.

[2] L. Karavidas, H. Apostolidis, dan T. Tsiatsos, "Usability Evaluation of an Adaptive Serious Game Prototype Based on Affective Feedback," *Information*, vol. 13, no. 9, hlm. 425, Sep 2022, doi: 10.3390/info13090425.

[3] E. A. Romero-Mendez, P. C. Santana-Mancilla, M. Garcia-Ruiz, O. A. Montesinos-López, dan L. E. Anido-Rifón, "The Use of Deep Learning to Improve Player Engagement in a Video Game through a Dynamic Difficulty Adjustment Based on Skills Classification," *Applied Sciences*, vol. 13, no. 14, hlm. 8249, Jul 2023, doi: 10.3390/app13148249.

[4] D. A. Alvarado Villa, O. A. Montesinos López, dan P. C. Santana-Mancilla, "Training of an intelligent agent to improve the gaming experience for video gamers," *Avances en IHC*, vol. 6, no. 1, hlm. 123–125, Nov 2021, doi: 10.47756/aihc.y6i1.105.

[5] I. S. W. B. Prasetya *dkk.*, "Programming Smart Playtesting," *ACM Trans. Softw. Eng. Methodol.*, vol. 35, no. 3, hlm. 1–47, Mar 2026, doi: 10.1145/3742473.

[6] Y. Zhao dan C. Tang, "Towards LLM-Based Automatic Playtest," dalam *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, Clarion Hotel Trondheim Trondheim Norway: ACM, Jun 2025, hlm. 1645–1651. doi: 10.1145/3696630.3728700.

[7] Y. Francillette, H. Tremblay, B. Bouchard, dan B.-A. Menelas, "A Comprehensive Model of Automated Evaluation of Difficulty in Platformer Games," *ACM Games: Research and Practice*, vol. 3, no. 1, Jan 2025, doi: 10.1145/3705013.

[8] Q. Delfosse *dkk.*, "Deep Reinforcement Learning Agents are not even close to Human Intelligence," 2025.

[9] X. Feng *dkk.*, "ChessGPT: Bridging Policy Learning and Language Modeling".

[10] S. Hu, T. Huang, dan L. Liu, "PokeLLMon: A Human-Parity Agent for Pokemon Battles with Large Language Models," *ArXiv*, vol. abs/2402.01118, 2024, [Daring]. Tersedia pada: https://api.semanticscholar.org/CorpusID:267406392

[11] G. Wang *dkk.*, "Voyager: An Open-Ended Embodied Agent with Large Language Models," 19 Oktober 2023, *arXiv*: arXiv:2305.16291. doi: 10.48550/arXiv.2305.16291.

[12] C. Xiao dan Z. Yang, "LLMs May Not Be Human-Level Players, But They Can Be Testers: Measuring Game Difficulty with LLM Agents," *Proc. ACM Hum.-Comput. Interact.*, vol. 9, no. 6, hlm. 1097–1123, Okt 2025, doi: 10.1145/3748634.

[13] K. Sreedhar dan L. Chilton, "Simulating Strategic Reasoning: Comparing the Ability of Single LLMs and Multi-Agent Systems to Replicate Human Behavior," dipresentasikan pada Hawaii International Conference on System Sciences, 2025. doi: 10.24251/HICSS.2025.100.

[14] T. Guzsvinecz dan J. Szűcs, "Game Theory and Strategic Decision-Making in the Dark Souls Games," dalam *Game Theory - Computational Aspects and Applications*, T. Guzsvinecz dan J. Szűcs, Ed., London: IntechOpen, 2024. doi: 10.5772/intechopen.1008456.

[15] P. Klepek, "'Nioh' Suggests That 'Dark Souls' Actually Invented a New Genre," Vice Magazine, 10 Februari 2017. [Daring]. Tersedia pada: https://www.vice.com/en/article/nioh-suggests-that-dark-souls-actually-invented-a-new-genre/

[16] T. Constant, G. Levieux, A. Buendia, dan S. Natkin, "From Objective to Subjective Difficulty Evaluation in Video Games," dalam *Human-Computer Interaction - INTERACT 2017*, vol. 10514, R. Bernhaupt, G. Dalvi, A. Joshi, D. K. Balkrishan, J. O'Neill, dan M. Winckler, Ed., dalam Lecture Notes in Computer Science, vol. 10514. , Cham: Springer International Publishing, 2017, hlm. 107–127. doi: 10.1007/978-3-319-67684-5_8.

[17] B. Bateni dan J. Whitehead, "Language-Driven Play: Large Language Models as Game-Playing Agents in Slay the Spire," dalam *Proceedings of the 19th International Conference on the*

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Foundations of Digital Games, Worcester MA USA: ACM, Mei 2024, hlm. 1–10. doi: 10.1145/3649921.3650013.

[18] T. Liang *dkk.*, "Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate," dalam *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Miami, Florida, USA: Association for Computational Linguistics, 2024, hlm. 17889–17904. doi: 10.18653/v1/2024.emnlp-main.992.